**CS 215: Task 01 1: Due: 08-Aug Time: as mentioned in Piazza. No extensions.**

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read the data correctly, we might be able to.
- Submission Guidelines. See `piazza` for standard static instructions.
- Your submission folder after extraction should look something like:

```
23B0019_task01/
├── readme.txt
└── task01.pdf
```

  You should probably name your zip file correctly. Notice how, AFTER extraction, the two files are inside the folder containing the roll number.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. A red die, a blue die, and a yellow die (all fair six-sided dice) are rolled. Let $B, R$, and $Y$ denote the numbers on the blue, red and yellow dice respectively.

   We are interested in the event $B < Y < R$.

   (a) What is the probability that no two dice land on the same number?

   Answer:

   There are $6^3 = 216$ equally likely outcomes.

   For all three numbers to be distinct,

   $$6 \times 5 \times 4 = 120$$

   ordered outcomes are possible.

   Thus,

   $$P(\text{all distinct}) = \frac{120}{216} = \frac{5}{9}.$$

   Marking scheme: 3 marks.

   (b) Given that no two dice land on the same number, what is the conditional probability that $B < Y < R$?

   Answer:

   Conditioned on distinct values, the six possible orderings of the three dice are equally likely:

   $$BYR,\ BRY,\ YBR,\ YRB,\ RBY,\ RYB.$$

   Only one ordering satisfies

   $$B < Y < R.$$

   Hence,

$$P(B < Y < R \mid \text{distinct}) = \frac{1}{6}.$$

Marking scheme: 3 marks.

(c) Find $P(B < Y < R)$ using part (a).

Answer:

Using part (a),

$$P(B < Y < R) = P(B < Y < R \mid \text{distinct})P(\text{distinct}).$$

Therefore,

$$P(B < Y < R) = \frac{1}{6} \cdot \frac{5}{9} = \boxed{\frac{5}{54}}.$$

Marking scheme: 3 marks.

(d) If we regard the outcome as the ordered vector $(B, R, Y)$, how many outcomes are there in the sample space?

Answer:

Each die has six possible outcomes.

Hence,

$$6 \times 6 \times 6 = 216.$$

Marking scheme: 2 marks.

(e) Without using the answer to part (c), can you determine the number of outcomes satisfying $B < Y < R$ in an alternate way?

Answer:

Choose any three distinct numbers from $\{1, \ldots, 6\}$.

The number of choices is

$$\binom{6}{3} = 20.$$

For each choice, exactly one assignment satisfies

$$B < Y < R.$$

Hence,

$$20$$

outcomes satisfy the required ordering.

Marking scheme: 4 marks.

(f) Use the results of parts (d) and (e) to verify the answer to part (c).

Answer:

Using parts (d) and (e),

$$P(B < Y < R) = \frac{20}{216} = \frac{5}{54},$$

which agrees with the answer obtained in part (c).

Marking scheme: 2 marks.

2. The sample mean and sample variance of five data values are, respectively,

$$\bar{x} = 104, \qquad s^2 = 16.$$

If three of the data values are $102, 100$, and $106$, determine the other two data values.

Answer:

Let the unknown observations be $x$ and $y$.

Since the sample mean is 104,

$$\frac{102 + 100 + 106 + x + y}{5} = 104,$$

which gives

$$x + y = 212.$$

Since the sample variance is

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1} = 16,$$

we have

$$\sum(x_i - 104)^2 = 4(16) = 64.$$

Hence,

$$(x - 104)^2 + (y - 104)^2 = 64 - 24 = 40.$$

Solving the above equations, we get the following solutions

Thus,

$$(x, y) = (110, 102)$$

or

$$(x, y) = (102, 110).$$

Therefore, the two missing observations are

$$\boxed{102 \text{ and } 110.}$$

Marking scheme (6 marks):

- Mean equation: 2 marks
- Variance equation: 2 marks
- Solving quadratic: 1 mark
- Correct answer: 1 mark

3. A computationally efficient way to compute the sample mean and sample variance is given by

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

and

$$s^2_{j+1} = \left(1 - \frac{1}{j}\right) s^2_j + (j+1)(\bar{x}_{j+1} - \bar{x}_j)^2.$$

(a) Use these formulas to compute the sample mean and sample variance of

$$3,\ 4,\ 7,\ 2,\ 9,\ 6.$$

Answer:
The computations are summarized below.

| $j$ | $\bar{x}_j$ | $s^2_j$ |
|---|---|---|
| 1 | 3 | 0 |
| 2 | 3.5 | 0.5 |
| 3 | $\frac{14}{3}$ | $\frac{13}{3}$ |
| 4 | 4 | $\frac{14}{3}$ |
| 5 | 5 | 8.5 |
| 6 | $\frac{31}{6}$ | $\frac{86}{15}$ |

Hence,

$$\boxed{\bar{x} = \frac{31}{6}, \qquad s^2 = \frac{86}{15}.}$$

Marking scheme (6 marks):

- Correct recursive means: 3 marks
- Correct recursive variances: 2 marks
- Final answer: 1 mark

(b) Verify your results in part (a) using the usual formulas for sample mean and sample variance.
Answer:
The sample mean is

$$\bar{x} = \frac{3+4+7+2+9+6}{6} = \frac{31}{6}.$$

Next,

$$\sum (x_i - \bar{x})^2 = \frac{172}{6} = \frac{86}{3}.$$

Therefore,

$$s^2 = \frac{1}{6-1}\left(\frac{86}{3}\right) = \frac{86}{15}.$$

Thus the values agree with those obtained in part (a).
Marking scheme (4 marks):

- Mean computation: 1 mark
- Variance computation: 2 marks
- Verification: 1 mark

(c) Verify the recursive formula

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1}.$$

Answer:
By definition,

$$\bar{x}_{j+1} = \frac{x_1 + \cdots + x_j + x_{j+1}}{j+1}.$$

Since

$$x_1 + \cdots + x_j = j\bar{x}_j,$$

we obtain

$$\bar{x}_{j+1} = \frac{j\bar{x}_j + x_{j+1}}{j+1}.$$

Rearranging,

$$\bar{x}_{j+1} = \bar{x}_j + \frac{x_{j+1} - \bar{x}_j}{j+1},$$

which proves the desired identity.
Marking scheme (4 marks):

- Start from definition: 2 marks
- Algebraic manipulation: 1 mark
- Final identity: 1 mark

4. An engineering system consisting of $n$ independent components is said to be a $k$-out-of-$n$ system if it functions whenever at least $k$ components function.

(a) If the $i^{\text{th}}$ component functions with probability $P_i$, $i = 1, 2, 3, 4$, compute the probability that a 2-out-of-4 system functions.
Answer:
The system functions if at least two of the four components function. Thus we consider the cases of exactly two, exactly three, and exactly four functioning components.

**Exactly two components function:**

$$\sum_{i=1}^{3} \sum_{j=i+1}^{4} P_i P_j \prod_{\substack{k=1 \\ k \neq i,j}}^{4} (1 - P_k).$$

**Exactly three components function:**

$$\sum_{i=1}^{4}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{4}P_j.$$

**All four components function:**

$$\prod_{i=1}^{4}P_i.$$

Hence,

$$\boxed{P(\text{system functions}) = \sum_{i=1}^{3}\sum_{j=i+1}^{4}P_iP_j\prod_{\substack{k=1\\ k\neq i,j}}^{4}(1-P_k) + \sum_{i=1}^{4}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{4}P_j + \prod_{i=1}^{4}P_i.}$$

Marking scheme (6 marks):

- Exactly two components: 3 marks
- Exactly three components: 2 marks
- All four components: 1 mark

(b) Generalize your reasoning from part (a) to write a compact expression for the probability that a 3-out-of-5 system functions. Use summation or subset notation; do not list all outcomes individually.

Answer:

The system functions if at least three of the five components function. Hence we consider the cases of exactly three, exactly four, and exactly five functioning components.

**Exactly three components function:**

$$\sum_{i=1}^{3}\sum_{j=i+1}^{4}\sum_{k=j+1}^{5}P_iP_jP_k\prod_{\substack{\ell=1\\ \ell\neq i,j,k}}^{5}(1-P_\ell).$$

**Exactly four components function:**

$$\sum_{i=1}^{5}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{5}P_j.$$

**All five components function:**

$$\prod_{i=1}^{5}P_i.$$

Therefore,

$$\boxed{\begin{aligned}P(\text{system functions}) = &\sum_{i=1}^{3}\sum_{j=i+1}^{4}\sum_{k=j+1}^{5}P_iP_jP_k\prod_{\substack{\ell=1\\ \ell\neq i,j,k}}^{5}(1-P_\ell)\\ &+\sum_{i=1}^{5}(1-P_i)\prod_{\substack{j=1\\ j\neq i}}^{5}P_j + \prod_{i=1}^{5}P_i.\end{aligned}}$$

Marking scheme (6 marks):

- Exactly three components: 3 marks
- Exactly four components: 2 marks
- All five components: 1 mark